

Unit 4. Regression Diagnostics

1 The Iterative Modelling Process

FREES §5.1.

Remark. Fitting is one step of a loop: specify a model, fit it, criticise the fit, respecify. The criticism step is this unit. Each of the classical assumptions of Unit 2 has a diagnostic attached to it, and a different remedy:

Assumption	Diagnostic	Section
A1 linearity	residuals against fitted values	§2
A3 constant variance	residual spread against fitted values; Breusch–Pagan	§5
A3 uncorrelated errors	residuals against time or index	Unit 9
A4 full rank	VIF, condition number	§4
A5 normality	normal quantile plot of residuals	§2
(none)	leverage, Cook’s distance	§3

The last row has no assumption attached: an influential observation violates nothing, but a conclusion resting on one observation is worth knowing about.

2 Residual Analysis

FREES §5.3 · ISLR §3.3.3.

2.1 Standardised and studentised residuals

By the residual-variance theorem of Unit 2, $\text{var}(e_i) = \sigma^2(1 - h_{ii})$: the raw residuals have unequal variances even under [A3](#), so they are not comparable across observations until rescaled.

Definition 4.1 (Standardised and studentised residuals). With $s^2 = \text{SSE}/(n - k - 1)$ and $s_{(i)}^2$ the same quantity computed with observation i deleted,

$$r_i = \frac{e_i}{s\sqrt{1 - h_{ii}}} \quad (\text{standardised}), \quad t_i = \frac{e_i}{s_{(i)}\sqrt{1 - h_{ii}}} \quad (\text{studentised, or deleted}). \quad (1)$$

Theorem 4.2 (The deleted variance without refitting).

$$s_{(i)}^2 = \frac{(n - k - 1)s^2 - e_i^2/(1 - h_{ii})}{n - k - 2}. \quad (2)$$

Proof. By the deletion identity of Lemma 4.3 below, $y_i - \mathbf{x}_i' \hat{\boldsymbol{\beta}}^{(-i)} = e_i / (1 - h_{ii})$, and the residual sum of squares of the deleted fit satisfies $\text{SSE}_{(i)} = \text{SSE} - e_i^2 / (1 - h_{ii})$: removing observation i removes its contribution to SSE, inflated by the same factor that inflates its prediction error. Dividing by its degrees of freedom $n - 1 - k - 1 = n - k - 2$ gives (2). $\square$

Remark. t_i has a t_{n-k-2} distribution under A1–A5, whereas r_i does not have a t distribution at all: its numerator and denominator both contain e_i , so an observation with a large residual inflates s and partly hides itself. Deleting the observation from the variance estimate removes that feedback, which is why t_i is the statistic for testing whether observation i is an outlier. The conventional flag is $|r_i| > 2$.

2.2 What to plot

Remark. Residuals are plotted against **fitted values**, not against the response: by the residual identities of Unit 2, $\hat{\mathbf{Y}}'\mathbf{e} = 0$ but $\mathbf{Y}'\mathbf{e} = \text{SSE} \neq 0$, so a plot against y_i shows a spurious upward trend in every data set.

- Curvature in residuals against fitted values indicates a violation of A1: the remedy is a trans-

formation or an added term (Unit 3 §3.4).

- A fan or funnel indicates non-constant variance, §5.
- A normal quantile plot bearing on A5 matters only for exact small-sample inference; in large samples the central limit theorem carries the t and F statistics regardless.

3 Influential Observations

FREES §5.4 · ISLR §3.3.3.

Remark (Three different things). An **outlier** has an unusual response given its predictors: large $|t_i|$. A **high-leverage** point has unusual predictors, irrespective of its response: large h_{ii} . An **influential** point is one whose removal materially changes the fit, which requires both. A high-leverage point that happens to lie on the fitted surface is not influential.

Remark (Leverage thresholds). h_{ii} averages $(k+1)/n$ by the leverage theorem of Unit 2. FREES declares high leverage at $h_{ii} > 3(k+1)/n$; ISLR uses $2(k+1)/n$. Both are conventions, not tests.

Lemma 4.3 (Deletion identity). Let $\hat{\boldsymbol{\beta}}^{(-i)}$ be the least squares estimate with observation i removed. Then

$$\hat{\boldsymbol{\beta}} - \hat{\boldsymbol{\beta}}^{(-i)} = \frac{(\mathbf{X}'\mathbf{X})^{-1}\mathbf{x}_i e_i}{1 - h_{ii}}, \quad y_i - \mathbf{x}_i' \hat{\boldsymbol{\beta}}^{(-i)} = \frac{e_i}{1 - h_{ii}}. \quad (3)$$

Proof. This is the deletion identity of Unit 1, proved there from the Sherman–Morrison formula applied to $\mathbf{X}'\mathbf{X} - \mathbf{x}_i\mathbf{x}_i'$. The second identity follows from the first on premultiplying by $\mathbf{x}_i'$ and subtracting from y_i . $\square$

Definition 4.4 (Cook's distance).

$$D_i = \frac{(\hat{\boldsymbol{\beta}} - \hat{\boldsymbol{\beta}}^{(-i)})' \mathbf{X}'\mathbf{X} (\hat{\boldsymbol{\beta}} - \hat{\boldsymbol{\beta}}^{(-i)})}{(k+1)s^2} = \frac{\sum_{j=1}^n (\hat{y}_j - \hat{y}_j^{(-i)})^2}{(k+1)s^2}. \quad (4)$$

Theorem 4.5 (Computable form of Cook's distance).

$$D_i = \frac{e_i^2}{(k+1)s^2} \cdot \frac{h_{ii}}{(1-h_{ii})^2} = \frac{r_i^2}{k+1} \cdot \frac{h_{ii}}{1-h_{ii}}. \quad (5)$$

Proof. Substituting (3) into (4),

$$(\hat{\beta} - \hat{\beta}^{(-i)})' \mathbf{X}' \mathbf{X} (\hat{\beta} - \hat{\beta}^{(-i)}) = \frac{e_i^2}{(1-h_{ii})^2} \mathbf{x}_i' (\mathbf{X}' \mathbf{X})^{-1} \mathbf{X}' \mathbf{X} (\mathbf{X}' \mathbf{X})^{-1} \mathbf{x}_i = \frac{e_i^2 h_{ii}}{(1-h_{ii})^2},$$

since $\mathbf{x}_i' (\mathbf{X}' \mathbf{X})^{-1} \mathbf{x}_i = h_{ii}$. Dividing by $(k+1)s^2$ gives the first form; substituting $e_i^2 = r_i^2 s^2 (1-h_{ii})$ gives the second. $\square$

Remark. The second form separates the two ingredients: r_i^2 measures how unusual the response is and $h_{ii}/(1-h_{ii})$ how unusual the predictors are. D_i is large only when both are appreciable, which is the formal content of the distinction above. A common flag is $D_i > 1$, or $D_i > 4/n$ for a more sensitive screen.

Definition 4.6 (DFFITS and DFBETAS).

$$\text{DFFITS}_i = \frac{\hat{y}_i - \hat{y}_i^{(-i)}}{s_{(i)}\sqrt{h_{ii}}}, \quad \text{DFBETAS}_{j,i} = \frac{\hat{\beta}_j - \hat{\beta}_j^{(-i)}}{s_{(i)}\sqrt{[(\mathbf{X}'\mathbf{X})^{-1}]_{jj}}}. \quad (6)$$

Proposition 4.7. $\text{DFFITS}_i = t_i\sqrt{h_{ii}/(1 - h_{ii})}$.

Proof. By (3), $\hat{y}_i - \hat{y}_i^{(-i)} = \mathbf{x}'_i(\hat{\boldsymbol{\beta}} - \hat{\boldsymbol{\beta}}^{(-i)}) = h_{ii}e_i/(1 - h_{ii})$. Dividing by $s_{(i)}\sqrt{h_{ii}}$,

$$\text{DFFITS}_i = \frac{h_{ii} e_i}{(1 - h_{ii}) s_{(i)}\sqrt{h_{ii}}} = \frac{e_i}{s_{(i)}\sqrt{1 - h_{ii}}} \sqrt{\frac{h_{ii}}{1 - h_{ii}}} = t_i\sqrt{\frac{h_{ii}}{1 - h_{ii}}}. \quad \square$$

Remark. D_i and DFFITS_i measure the same thing on different scales, one using s and the other $s_{(i)}$; they rank observations almost identically. DFBETAS is the finer instrument, reporting which **coefficient** an observation moves, at the cost of $k + 1$ numbers per observation. A conventional flag is $|\text{DFFITS}_i| > 2\sqrt{(k + 1)/n}$.

4 Collinearity

FREES §5.5.

Definition 4.8 (Variance inflation factor). Let R_j^2 be the coefficient of determination from regressing x_j on the other $k - 1$ explanatory variables and an intercept. Then

$$\text{VIF}_j = \frac{1}{1 - R_j^2}. \quad (7)$$

Theorem 4.9 (VIF and the standard error). For $j = 1, \dots, k$,

$$\text{SE}(\hat{\beta}_j) = \frac{s}{s_{x_j} \sqrt{n - 1}} \sqrt{\text{VIF}_j}. \quad (8)$$

Proof. By the Frisch–Waugh–Lovell theorem of Unit 2, applied with $\mathbf{X}_1 = \mathbf{x}_j$ and $\mathbf{X}_2$ the re-

remaining columns, $\hat{\beta}_j$ is the slope from regressing $\mathbf{Y}$ on $\mathbf{M}_2\mathbf{x}_j$, so

$$\text{var}(\hat{\beta}_j) = \frac{\sigma^2}{\mathbf{x}_j'\mathbf{M}_2\mathbf{x}_j}.$$

Now $\mathbf{x}_j'\mathbf{M}_2\mathbf{x}_j$ is the residual sum of squares from regressing x_j on the other variables, which is $(1 - R_j^2)$ times its total sum of squares $(n - 1)s_{x_j}^2$. Hence

$$\text{var}(\hat{\beta}_j) = \frac{\sigma^2}{(n - 1)s_{x_j}^2(1 - R_j^2)} = \frac{\sigma^2}{(n - 1)s_{x_j}^2} \text{VIF}_j,$$

and replacing σ by s and taking the square root gives (8). □

Remark. $\text{VIF}_j \geq 1$, with equality iff x_j is orthogonal to the other explanatory variables; it is the factor by which collinearity multiplies the **variance** of $\hat{\beta}_j$ over that orthogonal ideal. The usual flag is $\text{VIF}_j > 10$, equivalently $R_j^2 > 0.9$. Note that (8) contains no VIF for the intercept, and that VIF is unaffected by the response: collinearity is a property of $\mathbf{X}$.

Remark (What collinearity does and does not do). Collinearity does not bias $\hat{\beta}$, does not violate any of A1–A5, and does not damage prediction at predictor combinations resembling those observed. What it does is inflate standard errors, so that individually insignificant coefficients coexist with a highly significant joint F (Unit 3 §4), and make estimates unstable across samples. The remedies are to drop one of the redundant variables, to combine them, or to shrink; ridge regression (Unit 6) is the direct attack, replacing $(\mathbf{X}'\mathbf{X})^{-1}$ by $(\mathbf{X}'\mathbf{X} + \lambda\mathbf{I})^{-1}$.

5 Heteroscedasticity

FREES §5.7.

Assume A1, A2 and A4, but replace A3 by $\text{var}(\boldsymbol{\varepsilon}) = \boldsymbol{\Omega} = \text{diag}(\sigma_1^2, \dots, \sigma_n^2)$.

Theorem 4.10 (OLS under heteroscedasticity). The least squares estimator remains unbiased, but

$$\text{var}(\hat{\beta}) = (\mathbf{X}'\mathbf{X})^{-1} \left(\sum_{i=1}^n \sigma_i^2 \mathbf{x}_i \mathbf{x}_i' \right) (\mathbf{X}'\mathbf{X})^{-1} \neq \sigma^2 (\mathbf{X}'\mathbf{X})^{-1}, \quad (9)$$

and $\hat{\boldsymbol{\beta}}$ is no longer best linear unbiased.

Proof. Unbiasedness used only A1–A2, so it survives. For the variance, apply $\text{var}(\mathbf{A}\mathbf{z}) = \mathbf{A} \text{var}(\mathbf{z}) \mathbf{A}'$ to $\hat{\boldsymbol{\beta}} - \boldsymbol{\beta} = (\mathbf{X}'\mathbf{X})^{-1}\mathbf{X}'\boldsymbol{\varepsilon}$:

$$\text{var}(\hat{\boldsymbol{\beta}}) = (\mathbf{X}'\mathbf{X})^{-1}\mathbf{X}'\boldsymbol{\Omega}\mathbf{X}(\mathbf{X}'\mathbf{X})^{-1},$$

and $\mathbf{X}'\boldsymbol{\Omega}\mathbf{X} = \sum_i \sigma_i^2 \mathbf{x}_i \mathbf{x}_i'$. The Gauss–Markov proof used $\text{var}(\boldsymbol{\varepsilon}) = \sigma^2 \mathbf{I}$ at the step $\text{var}(\tilde{\boldsymbol{\beta}}) = \sigma^2 \mathbf{C}\mathbf{C}'$, so its conclusion no longer follows; Theorem 4.12 exhibits an unbiased linear estimator with smaller variance. $\square$

Remark. The reported standard errors, built from $s^2(\mathbf{X}'\mathbf{X})^{-1}$, estimate the wrong quantity. They may be too small or too large: which, depends on whether the large variances sit at high-leverage or low-leverage points.

Definition 4.11 (White’s heteroscedasticity-consistent estimator). Replacing the unknown σ_i^2

in (9) by the squared residuals,

$$\widehat{\text{var}}(\hat{\boldsymbol{\beta}}) = (\mathbf{X}'\mathbf{X})^{-1} \left(\sum_{i=1}^n c e_i^2 \mathbf{x}_i \mathbf{x}_i' \right) (\mathbf{X}'\mathbf{X})^{-1}, \quad (10)$$

with $c = 1$ for HC0 and $c = n/(n - k - 1)$ for the small-sample-corrected HC1.

Remark. The coefficients are untouched: only the standard errors change. This is the cheapest response to heteroscedasticity of unknown form, and requires no model for σ_i^2 .

Theorem 4.12 (Weighted least squares is best linear unbiased). Define

$$\mathbf{W} = \boldsymbol{\Omega}^{-1} = \text{diag}(1/\sigma_1^2, \dots, 1/\sigma_n^2)$$

and

$$\hat{\boldsymbol{\beta}}_W = (\mathbf{X}'\mathbf{W}\mathbf{X})^{-1} \mathbf{X}'\mathbf{W}\mathbf{Y}, \quad (11)$$

the minimiser of $\sum_i w_i (y_i - \mathbf{x}_i' \boldsymbol{\beta})^2$ with $w_i = 1/\sigma_i^2$. Then $\hat{\boldsymbol{\beta}}_W$ is unbiased with $\text{var}(\hat{\boldsymbol{\beta}}_W) = (\mathbf{X}'\mathbf{W}\mathbf{X})^{-1}$, and it is best linear unbiased.

Proof. Put $\mathbf{W}^{1/2} = \text{diag}(1/\sigma_i)$ and transform to $\mathbf{Y}^* = \mathbf{W}^{1/2}\mathbf{Y}$, $\mathbf{X}^* = \mathbf{W}^{1/2}\mathbf{X}$, $\boldsymbol{\varepsilon}^* = \mathbf{W}^{1/2}\boldsymbol{\varepsilon}$. Then

$$\text{var}(\boldsymbol{\varepsilon}^*) = \mathbf{W}^{1/2}\boldsymbol{\Omega}\mathbf{W}^{1/2} = \mathbf{I}_n,$$

so the starred model satisfies A1–A4 with $\sigma^2 = 1$. Ordinary least squares on it is

$$(\mathbf{X}^{*\prime}\mathbf{X}^*)^{-1}\mathbf{X}^{*\prime}\mathbf{Y}^* = (\mathbf{X}'\mathbf{W}\mathbf{X})^{-1}\mathbf{X}'\mathbf{W}\mathbf{Y} = \hat{\boldsymbol{\beta}}_W,$$

which by the Gauss–Markov theorem is best linear unbiased among estimators linear in $\mathbf{Y}^*$, and hence in $\mathbf{Y}$, the map $\mathbf{Y} \mapsto \mathbf{Y}^*$ being linear and invertible. Its variance is $1 \cdot (\mathbf{X}^{*\prime}\mathbf{X}^*)^{-1} = (\mathbf{X}'\mathbf{W}\mathbf{X})^{-1}$. □

Remark. Weighted least squares therefore **is** ordinary least squares, on data rescaled so that every observation has unit error variance: an observation with large σ_i^2 is down-weighted because it carries less information. The practical obstacle is that σ_i^2 is unknown; it is usually modelled as a function of the predictors or of the fitted value, or sidestepped by transforming the response so that the variance is approximately constant.

5.1 Detecting heteroscedasticity

Plot the residuals against the fitted values and look for a fan. The Breusch–Pagan test formalises this.

Definition 4.13 (Variance model and hypothesis). Let $\mathbf{Y} = \mathbf{X}\boldsymbol{\beta} + \boldsymbol{\varepsilon}$ with ε_i independent $N(0, \sigma_i^2)$ and

$$\sigma_i^2 = h(\alpha_0 + \mathbf{z}_i' \boldsymbol{\gamma}) = h(\mathbf{w}_i' \boldsymbol{\alpha}), \quad \mathbf{w}_i = \begin{pmatrix} 1 \\ \mathbf{z}_i \end{pmatrix}, \quad \boldsymbol{\alpha} = \begin{pmatrix} \alpha_0 \\ \boldsymbol{\gamma} \end{pmatrix}, \quad (12)$$

where $\mathbf{z}_i$ is a known p -vector, $h > 0$ is differentiable and $h'(\alpha_0) \neq 0$. The null hypothesis is $H_0: \boldsymbol{\gamma} = \mathbf{0}$, under which $\sigma_i^2 = h(\alpha_0)$ does not vary with i . Write $\mathbf{Z}$ for the $n \times p$ matrix with rows $\mathbf{z}_i'$, $\mathbf{W} = [\mathbf{1} \ \mathbf{Z}]$, and $\mathbf{M}_0 = \mathbf{I}_n - n^{-1} \mathbf{1} \mathbf{1}'$.

Lemma 4.14 (Score for the variance parameters). With $u_i = y_i - \mathbf{x}_i' \boldsymbol{\beta}$, the log-likelihood

$$\ell(\boldsymbol{\beta}, \boldsymbol{\alpha}) = -\frac{n}{2} \ln(2\pi) - \frac{1}{2} \sum_{i=1}^n \ln \sigma_i^2 - \frac{1}{2} \sum_{i=1}^n \frac{u_i^2}{\sigma_i^2} \quad (13)$$

has

$$\frac{\partial \ell}{\partial \boldsymbol{\alpha}} = \frac{1}{2} \sum_{i=1}^n \frac{h'(\mathbf{w}_i' \boldsymbol{\alpha})}{\sigma_i^2} \left(\frac{u_i^2}{\sigma_i^2} - 1 \right) \mathbf{w}_i. \quad (14)$$

Proof. Since $\sigma_i^2 = h(\mathbf{w}_i' \boldsymbol{\alpha})$, the chain rule gives $\partial \sigma_i^2 / \partial \boldsymbol{\alpha} = h'(\mathbf{w}_i' \boldsymbol{\alpha}) \mathbf{w}_i$, and therefore $\partial(\sigma_i^{-2}) / \partial \boldsymbol{\alpha} = -\sigma_i^{-4} h'(\mathbf{w}_i' \boldsymbol{\alpha}) \mathbf{w}_i$. Differentiating the two sums in (13) term by term,

$$\frac{\partial}{\partial \boldsymbol{\alpha}} \left(-\frac{1}{2} \ln \sigma_i^2 \right) = -\frac{h'(\mathbf{w}_i' \boldsymbol{\alpha})}{2\sigma_i^2} \mathbf{w}_i, \quad \frac{\partial}{\partial \boldsymbol{\alpha}} \left(-\frac{1}{2} \frac{u_i^2}{\sigma_i^2} \right) = \frac{u_i^2 h'(\mathbf{w}_i' \boldsymbol{\alpha})}{2\sigma_i^4} \mathbf{w}_i,$$

and adding them gives the summand of (14). □

Lemma 4.15 (Fourth moment). If $Z \sim N(0, 1)$ then $\mathbf{E}[Z^4] = 3$; hence $\text{var}(Z^2) = 2$, and for $\varepsilon \sim N(0, \sigma^2)$, $\text{var}(\varepsilon^2 / \sigma^2) = 2$.

Proof. The standard normal density satisfies $\varphi'(z) = -z\varphi(z)$, so integrating by parts,

$$\mathbb{E}[Z^4] = \int z^3 \cdot z\varphi(z) \, dz = - \int z^3 \varphi'(z) \, dz = \left[-z^3 \varphi(z) \right]_{-\infty}^{\infty} + 3 \int z^2 \varphi(z) \, dz = 0 + 3,$$

the boundary term vanishing because φ decays faster than any polynomial grows. Then $\text{var}(Z^2) = \mathbb{E}[Z^4] - (\mathbb{E}[Z^2])^2 = 3 - 1 = 2$. $\square$

Lemma 4.16 (Covariance of the score). $\frac{\partial \ell}{\partial \boldsymbol{\alpha}}$ has covariance matrix

$$\mathbf{I}_{\boldsymbol{\alpha}} = \frac{1}{2} \sum_{i=1}^n \frac{h'(\mathbf{w}_i' \boldsymbol{\alpha})^2}{\sigma_i^4} \mathbf{w}_i \mathbf{w}_i', \quad (15)$$

and it is uncorrelated with $\partial \ell / \partial \boldsymbol{\beta}$.

Proof. The i -th summand of (14) is $(h'_i/2\sigma_i^2)(u_i^2/\sigma_i^2 - 1)\mathbf{w}_i$ with $h'_i = h'(\mathbf{w}_i' \boldsymbol{\alpha})$. By Lemma 4.15, $\text{var}(u_i^2/\sigma_i^2 - 1) = 2$, so that summand has covariance $(h_i'^2/4\sigma_i^4) \cdot 2 \cdot \mathbf{w}_i \mathbf{w}_i'$. The summands are

independent across i , so the covariances add, giving (15).

For the second claim, $\partial\ell/\partial\boldsymbol{\beta} = \sum_i \mathbf{x}_i u_i / \sigma_i^2$ is linear in the u_i while (14) is a function of the u_i^2 ; for a symmetric distribution $\mathbf{E}[u_i^3] = 0$, so $\text{cov}(u_i, u_i^2 / \sigma_i^2 - 1) = \mathbf{E}[u_i^3] / \sigma_i^2 = 0$. $\square$

Lemma 4.17 (Restricted maximum likelihood). Under H_0 the maximisers of (13) are $\tilde{\boldsymbol{\beta}} = \hat{\boldsymbol{\beta}}$, the least squares estimator, and $\tilde{\sigma}^2 = \text{SSE}/n$. Consequently, writing

$$g_i = \frac{e_i^2}{\tilde{\sigma}^2} - 1, \quad (16)$$

one has $\mathbf{1}'\mathbf{g} = 0$ and hence $\mathbf{Z}'\mathbf{g} = \mathbf{Z}'\mathbf{M}_0\mathbf{g}$.

Proof. Under H_0 all σ_i^2 equal a common σ^2 , so (13) is the homoscedastic normal log-likelihood, whose maximisers are $\hat{\boldsymbol{\beta}}$ and SSE/n by the maximum likelihood theorem of Unit 2. Then

$$\mathbf{1}'\mathbf{g} = \sum_{i=1}^n \left(\frac{e_i^2}{\tilde{\sigma}^2} - 1 \right) = \frac{\text{SSE}}{\text{SSE}/n} - n = 0,$$

and $\mathbf{Z}'\mathbf{M}_0\mathbf{g} = \mathbf{Z}'\mathbf{g} - n^{-1}(\mathbf{Z}'\mathbf{1})(\mathbf{1}'\mathbf{g}) = \mathbf{Z}'\mathbf{g}$. □

Remark. The identity $\mathbf{1}'\mathbf{g} = 0$ is the reason the scaling in (16) must be $\tilde{\sigma}^2 = \text{SSE}/n$ and not $s^2 = \text{SSE}/(n - k - 1)$: only the maximum likelihood estimator makes the score component along the intercept column of $\mathbf{W}$ vanish, which is what the next theorem needs.

Definition 4.18 (Score statistic). Evaluate the score at the restricted maximiser. Its $\boldsymbol{\beta}$ -component and its α_0 -component are then zero, the first because $\tilde{\boldsymbol{\beta}}$ maximises over $\boldsymbol{\beta}$, the second by Lemma 4.17, so only $\mathbf{U}_\gamma$ survives, and the **score** (Lagrange multiplier) statistic is

$$LM = \mathbf{U}'_\gamma [(\mathbf{I}_\alpha^{-1})_{\gamma\gamma}]^{-1} \mathbf{U}_\gamma, \quad (17)$$

the middle factor being the inverse of the $\gamma\gamma$ block of $\mathbf{I}_\alpha^{-1}$, which by Lemma 4.16 is the relevant covariance once the nuisance parameters $\boldsymbol{\beta}$ and α_0 are accounted for.

Theorem 4.19 (The Breusch–Pagan statistic). Under H_0 , the statistic (17) equals

$$LM = \frac{1}{2} (\mathbf{Z}'\mathbf{g})'(\mathbf{Z}'\mathbf{M}_0\mathbf{Z})^{-1}(\mathbf{Z}'\mathbf{g}) = \frac{1}{2} \text{SSR}_z \quad (18)$$

where SSR_z is the regression sum of squares from regressing $\mathbf{g}$ (equivalently $e_i^2/\tilde{\sigma}^2$, or e_i^2 up to the scaling below) on $\mathbf{Z}$ with an intercept. In particular LM does not depend on h .

Proof. Step 1: the score. Under H_0 the variances are all $\tilde{\sigma}^2$ and $h'(\mathbf{w}'_i\tilde{\boldsymbol{\alpha}}) = h'(\tilde{\alpha}_0) =: h'_0$ for every i , so Lemma 4.14 gives

$$\mathbf{U}_\alpha = \frac{h'_0}{2\tilde{\sigma}^2} \sum_{i=1}^n g_i \mathbf{w}_i = \frac{h'_0}{2\tilde{\sigma}^2} \mathbf{W}'\mathbf{g}, \quad \text{hence} \quad \mathbf{U}_\gamma = \frac{h'_0}{2\tilde{\sigma}^2} \mathbf{Z}'\mathbf{g}.$$

Step 2: the covariance. By Lemma 4.16 with σ_i^2 and h'_i constant,

$$\mathbf{I}_\alpha = c \mathbf{W}'\mathbf{W}, \quad c = \frac{h_0'^2}{2\tilde{\sigma}^4}.$$

For a partitioned symmetric matrix the $\gamma\gamma$ block of the inverse is the inverse of the Schur complement, and the Schur complement of the leading $(1, 1)$ entry of $\mathbf{W}'\mathbf{W} = \begin{pmatrix} n & \mathbf{1}'\mathbf{Z} \\ \mathbf{Z}'\mathbf{1} & \mathbf{Z}'\mathbf{Z} \end{pmatrix}$ is

$$\mathbf{Z}'\mathbf{Z} - \mathbf{Z}'\mathbf{1} n^{-1} \mathbf{1}'\mathbf{Z} = \mathbf{Z}'\mathbf{M}_0\mathbf{Z}.$$

Hence $[(\mathbf{I}_\alpha^{-1})_{\gamma\gamma}]^{-1} = c \mathbf{Z}'\mathbf{M}_0\mathbf{Z}$, and

$$LM = \frac{1}{c} (\mathbf{Z}'\mathbf{g})' (\mathbf{Z}'\mathbf{M}_0\mathbf{Z})^{-1} (\mathbf{Z}'\mathbf{g}) \cdot \frac{h_0'^2}{4\tilde{\sigma}^4} = \frac{2\tilde{\sigma}^4}{h_0'^2} \cdot \frac{h_0'^2}{4\tilde{\sigma}^4} (\mathbf{Z}'\mathbf{g})' (\mathbf{Z}'\mathbf{M}_0\mathbf{Z})^{-1} (\mathbf{Z}'\mathbf{g}),$$

which is the first equality of (18). The factor $h_0'^2$ cancels, so the statistic is the same for every differentiable h with $h'(\alpha_0) \neq 0$: the test has power against a whole family of variance models without committing to one.

Step 3: identification with SSR_z . Regressing $\mathbf{g}$ on $\mathbf{W} = [\mathbf{1} \ \mathbf{Z}]$, the regression sum of squares is $\mathbf{g}'(\mathbf{H}_W - \mathbf{H}_0)\mathbf{g}$ with $\mathbf{H}_W$ the projection onto $\mathcal{C}(\mathbf{W})$ and $\mathbf{H}_0 = n^{-1}\mathbf{1}\mathbf{1}'$. By Frisch–Waugh–Lovell (Unit 3), $\mathbf{H}_W - \mathbf{H}_0 = \mathbf{M}_0\mathbf{Z}(\mathbf{Z}'\mathbf{M}_0\mathbf{Z})^{-1}\mathbf{Z}'\mathbf{M}_0$, so

$$\text{SSR}_z = (\mathbf{Z}'\mathbf{M}_0\mathbf{g})' (\mathbf{Z}'\mathbf{M}_0\mathbf{Z})^{-1} (\mathbf{Z}'\mathbf{M}_0\mathbf{g}) = (\mathbf{Z}'\mathbf{g})' (\mathbf{Z}'\mathbf{M}_0\mathbf{Z})^{-1} (\mathbf{Z}'\mathbf{g})$$

using $\mathbf{Z}'\mathbf{M}_0\mathbf{g} = \mathbf{Z}'\mathbf{g}$ from Lemma 4.17. Finally, replacing $\mathbf{g}$ by $e_i^2/\tilde{\sigma}^2 = g_i + 1$ changes the regressand by a constant vector, which is absorbed by the intercept and leaves SSR_z unchanged. $\square$

Theorem 4.20 (Null distribution). Assume H_0 , that $n^{-1}\mathbf{Z}'\mathbf{M}_0\mathbf{Z} \rightarrow \mathbf{Q}$ positive definite, that $\max_i h_{ii} \rightarrow 0$, and that the $\mathbf{z}_i$ satisfy a Lindeberg condition. Then

$$LM \xrightarrow{d} \chi_p^2. \quad (19)$$

Proof. Step 1: replace residuals by errors. $e_i = \varepsilon_i - \mathbf{x}_i'(\hat{\boldsymbol{\beta}} - \boldsymbol{\beta})$, and $\hat{\boldsymbol{\beta}} - \boldsymbol{\beta} = O_p(n^{-1/2})$, so $e_i^2 = \varepsilon_i^2 + O_p(n^{-1/2})$ uniformly under $\max_i h_{ii} \rightarrow 0$; likewise $\tilde{\sigma}^2 = \sigma^2 + O_p(n^{-1/2})$. Hence $n^{-1/2}\mathbf{Z}'\mathbf{g} = n^{-1/2}\sum_i \mathbf{z}_i(\varepsilon_i^2/\sigma^2 - 1) + o_p(1)$.

Step 2: central limit theorem. The summands $\mathbf{z}_i(\varepsilon_i^2/\sigma^2 - 1)$ are independent with mean zero, and by Lemma 4.15 each has covariance $2\mathbf{z}_i\mathbf{z}_i'$. Their average covariance is $2n^{-1}\mathbf{Z}'\mathbf{Z}$, and centring by the intercept replaces this by $2n^{-1}\mathbf{Z}'\mathbf{M}_0\mathbf{Z} \rightarrow 2\mathbf{Q}$. The Lindeberg–Feller central limit theorem gives

$$n^{-1/2}\mathbf{Z}'\mathbf{g} \xrightarrow{d} N(\mathbf{0}, 2\mathbf{Q}).$$

Step 3: the quadratic form. By (18),

$$LM = \frac{1}{2} (n^{-1/2} \mathbf{Z}' \mathbf{g})' (n^{-1} \mathbf{Z}' \mathbf{M}_0 \mathbf{Z})^{-1} (n^{-1/2} \mathbf{Z}' \mathbf{g}),$$

and $n^{-1} \mathbf{Z}' \mathbf{M}_0 \mathbf{Z} \rightarrow \mathbf{Q}$. If $\mathbf{V} \sim N(\mathbf{0}, 2\mathbf{Q})$ then $(2\mathbf{Q})^{-1/2} \mathbf{V} \sim N(\mathbf{0}, \mathbf{I}_p)$, so $\frac{1}{2} \mathbf{V}' \mathbf{Q}^{-1} \mathbf{V} = \|(2\mathbf{Q})^{-1/2} \mathbf{V}\|^2 \sim \chi_p^2$ by the chi-square theorem for quadratic forms of Unit 2. The continuous mapping theorem transfers this to the limit of LM . $\square$

Remark (Where the limit theory comes from). Step 2 uses the Lindeberg–Feller central limit theorem in its weighted form and Step 3 the continuous mapping theorem. Neither is assumed: the **Mathematical Prerequisites** note proves the Lindeberg–Feller theorem and its weighted-sums corollary, the continuous mapping theorem, Slutsky’s theorem, which is what permits σ^2 and the scale κ to be replaced by consistent estimates, and the theorem on quadratic forms of asymptotically normal vectors that converts $\mathbf{Z}' \mathbf{g}$ into a χ_p^2 limit. The only things taken as given anywhere in that chain are the completeness of $\mathbb{R}$ and the four convergence theorems of the Lebesgue integral.

Theorem 4.21 (Koenker's studentised statistic). Drop normality, retaining independence, $\mathbf{E}[\varepsilon_i] = 0$, $\text{var}(\varepsilon_i) = \sigma^2$ and $\tau^2 := \text{var}(\varepsilon_i^2) = \mathbf{E}[\varepsilon_i^4] - \sigma^4 < \infty$. Then $n^{-1/2}\mathbf{Z}'\mathbf{g} \xrightarrow{d} N(\mathbf{0}, (\tau^2/\sigma^4)\mathbf{Q})$, so (18) is misscaled by the factor $\tau^2/(2\sigma^4)$. Estimating τ^2 by $\hat{\tau}^2 = n^{-1} \sum_i (e_i^2 - \tilde{\sigma}^2)^2$ gives the correctly scaled statistic

$$LM_{\text{stud}} = \frac{\tilde{\sigma}^4}{\hat{\tau}^2} (\mathbf{Z}'\mathbf{g})'(\mathbf{Z}'\mathbf{M}_0\mathbf{Z})^{-1}(\mathbf{Z}'\mathbf{g}) = n R_z^2 \xrightarrow{d} \chi_p^2, \quad (20)$$

where R_z^2 is the coefficient of determination from regressing e_i^2 on $\mathbf{Z}$ with an intercept. Moreover

$$\frac{LM}{LM_{\text{stud}}} = \frac{\hat{\tau}^2}{2\tilde{\sigma}^4}. \quad (21)$$

Proof. Only Lemma 4.15 used normality, and only to assert $\text{var}(\varepsilon_i^2/\sigma^2) = 2$; in general that variance is τ^2/σ^4 . Repeating Step 2 of Theorem 4.20 with this value gives the stated limit, and the quadratic form standardised by σ^4/τ^2 rather than $1/2$ is χ_p^2 .

For the identification, put $\mathbf{v} = (e_1^2, \dots, e_n^2)'$, so that $\bar{v} = \tilde{\sigma}^2$ and $\mathbf{g} = \mathbf{M}_0\mathbf{v}/\tilde{\sigma}^2$. Since $(\mathbf{H}_W - \mathbf{H}_0)\mathbf{1} =$

0 we have $\mathbf{M}_0(\mathbf{H}_W - \mathbf{H}_0) = \mathbf{H}_W - \mathbf{H}_0$, hence

$$\text{SSR}_z = \mathbf{g}'(\mathbf{H}_W - \mathbf{H}_0)\mathbf{g} = \tilde{\sigma}^{-4} \mathbf{v}'\mathbf{M}_0(\mathbf{H}_W - \mathbf{H}_0)\mathbf{M}_0\mathbf{v} = \tilde{\sigma}^{-4} \mathbf{v}'(\mathbf{H}_W - \mathbf{H}_0)\mathbf{v} = \frac{\text{SSR}_v}{\tilde{\sigma}^4},$$

where SSR_v is the regression sum of squares from regressing $\mathbf{v}$ on $[\mathbf{1} \ \mathbf{Z}]$. Also $\hat{\tau}^2 = n^{-1} \sum_i (e_i^2 - \bar{v})^2 = \text{SST}_v/n$. Therefore

$$LM_{\text{stud}} = \frac{\tilde{\sigma}^4}{\hat{\tau}^2} \text{SSR}_z = \frac{\tilde{\sigma}^4}{\text{SST}_v/n} \cdot \frac{\text{SSR}_v}{\tilde{\sigma}^4} = n \frac{\text{SSR}_v}{\text{SST}_v} = nR_z^2,$$

and dividing $LM = \text{SSR}_z/2$ by this gives (21). □

Remark (Which one R computes, and what their ratio measures). `lmtest::bptest` returns LM_{stud} **by default**; the statistic of Theorem 4.19 is `bptest(m, studentize = FALSE)`. The two can differ substantially, so a hand computation following Theorem 4.19 compared against a bare `bptest` call will appear to disagree.

By (21) the ratio is $\hat{\tau}^2/(2\tilde{\sigma}^4)$, which estimates $\text{var}(\varepsilon^2)/\{2\text{var}(\varepsilon)^2\}$, which is exactly 1 when the

errors are normal, by Lemma 4.15. The ratio is therefore itself a measurement: how far the fourth moment of the errors departs from the normal value. §6.4 computes it.

6 Computation: Formulas Against Package Output

ISLR §3.6.3.

```
1 library(ISLR2)
2 fit <- lm(medv ~ lstat + age + rm, data = Boston)
3 X <- model.matrix(fit); y <- Boston$medv
4 n <- nrow(X); k <- ncol(X) - 1L
5 e <- residuals(fit); h <- hatvalues(fit)
6 s2 <- sum(e^2) / (n - k - 1)
7 c(n = n, k = k, df = n - k - 1)
```

```
##      n      k  df
## 506      3 502
```

6.1 Residuals (§2)

```
1 r_man <- e / (sqrt(s2) * sqrt(1 - h))           # standardised
2 s2i   <- ((n-k-1)*s2 - e^2/(1-h)) / (n-k-2)    # deleted variance
```

```

3 t_man <- e / (sqrt(s2i) * sqrt(1 - h))                                # studentised
4 c(rstandard = max(abs(r_man - rstandard(fit))),
5   rstudent  = max(abs(t_man - rstudent(fit))),
6   sd_raw     = sd(e), sd_standardised = sd(r_man))

```

```

##          rstandard          rstudent          sd_raw
##      0.000e+00      1.332e-15      5.526e+00
## sd_standardised
##      1.005e+00

```

The studentised residual exceeds the standardised one exactly where the observation inflates s :

```

1 i <- which.max(abs(e))
2 c(index = i, e_i = e[i], h_ii = h[i],
3   standardised = r_man[i], studentised = t_man[i],
4   s = sqrt(s2), s_deleted = sqrt(s2i[i]))

```

```

##          index.369          e_i.369          h_ii.369
##      369.00000      27.50044      0.04791
## standardised.369  studentised.369          s
##          5.08538          5.21645      5.54215
##      s_deleted.369

```

```
##          5.40289
```

```
1 c(flag_2 = sum(abs(r_man) > 2),
2   flag_3 = sum(abs(r_man) > 3),
3   expected_2_if_normal = round(2 * pnorm(-2) * n, 1))
```

```
##          flag_2          flag_3
##          28          9
## expected_2_if_normal
##          23
```

6.2 Influence (§3)

The deletion identity of Lemma 4.3, against n explicit refits:

```
1 b <- coef(fit); XtXi <- solve(t(X) %*% X)
2 idx <- c(1, 100, 215, 373, 506)
3 delta_man <- sapply(idx, function(i)
4   drop(XtXi %*% X[i, ] * e[i] / (1 - h[i])))
5 delta_ref <- sapply(idx, function(i)
6   b - coef(lm(medv ~ lstat + age + rm, data = Boston[-i, ])))
7 c(max_abs_diff = max(abs(delta_man - delta_ref)))
```

```
## max_abs_diff
##      6.528e-14
```

```
1 D_man <- e^2 / ((k+1) * s2) * h / (1 - h)^2           # first form
2 D_alt <- r_man^2 / (k+1) * h / (1 - h)                # second form
3 c(form1_vs_R = max(abs(D_man - cooks.distance(fit))),
4   form2_vs_R = max(abs(D_alt - cooks.distance(fit))),
5   max_D = max(D_man), n_above_4_over_n = sum(D_man > 4/n))
```

```
##          form1_vs_R          form2_vs_R          max_D
##          1.110e-16          1.110e-16          3.253e-01
## n_above_4_over_n
##          4.400e+01
```

```
1 dff_man <- t_man * sqrt(h / (1 - h))                  # Proposition
2 c(dffits_vs_R = max(abs(dff_man - dffits(fit))),
3   dfbetas_vs_R = max(abs(
4     (b["lstat"] - sapply(idx, function(i)
5       coef(lm(medv ~ lstat + age + rm, data = Boston[-i, ]))["lstat"]))) /
6     (sqrt(s2i[idx]) * sqrt(XtXi["lstat", "lstat"]))) -
7     dfbetas(fit)[idx, "lstat"])))
```

```
## dffits_vs_R dfbetas_vs_R
##      2.220e-16      1.565e-14
```

Outlier, leverage and influence rank different observations:

```
1 top <- function(v) order(abs(v), decreasing = TRUE)[1:5]
2 rbind(by_studentised = top(t_man),
3       by_leverage    = top(h),
4       by_cooks_D     = top(D_man))
```

```
##           [,1] [,2] [,3] [,4] [,5]
## by_studentised 369  373  372  370  375
## by_leverage    366  215  368  369  365
## by_cooks_D     369  366  215  375  365
```

6.3 Collinearity (§4)

```
1 suppressMessages(library(car))
2 Xn <- X[, -1, drop = FALSE]
3 vif_man <- sapply(seq_len(ncol(Xn)), function(j)
4   1 / (1 - summary(lm(Xn[, j] ~ Xn[, -j]))$r.squared)) # 1/(1 - R_j^2)
5 rbind(manual = vif_man, car = vif(fit))
```

```
##          lstat   age    rm
## manual 2.477 1.639 1.675
## car    2.477 1.639 1.675
```

Theorem 4.9 against the standard errors R reports:

```
1 sx <- apply(Xn, 2, sd)
2 se_from_vif <- sqrt(s2) / (sx * sqrt(n - 1)) * sqrt(vif_man)
3 rbind(from_VIF = se_from_vif,
4       lm       = coef(summary(fit))[-1, "Std. Error"])
```

```
##          lstat   age    rm
## from_VIF 0.05436 0.01122 0.4543
## lm       0.05436 0.01122 0.4543
```

Adding a nearly redundant column inflates the standard errors without touching the fit:

```
1 set.seed(1)
2 Boston$lstat2 <- Boston$lstat + rnorm(n, sd = 0.01)      # almost a copy
3 m_col <- lm(medv ~ lstat + lstat2 + age + rm, data = Boston)
4 rbind(
5   se = c(base = coef(summary(fit))["lstat", "Std. Error"],
6         with_copy = coef(summary(m_col))["lstat", "Std. Error"]),
```

```

7  t    = c(coef(summary(fit))["lstat", "t value"],
8          coef(summary(m_col))["lstat", "t value"]))

```

```

##          base with_copy
## se    0.05436    24.527
## t   -12.29846     0.673

```

```

1  c(R2_base = summary(fit)$r.squared, R2_col = summary(m_col)$r.squared,
2    VIF_lstat = vif(m_col)[["lstat"]])

```

```

##      R2_base      R2_col    VIF_lstat
##      0.6390      0.6394 503876.2292

```

6.4 Heteroscedasticity (§5)

A design where the truth is known: $\text{var}(\varepsilon_i)$ grows with x_i .

```

1  set.seed(2)
2  nh  <- 300
3  xh  <- runif(nh, 1, 10)
4  sig <- 0.5 * xh                                     # true sd_i
5  yh  <- 2 + 1.5 * xh + rnorm(nh, sd = sig)
6  mh  <- lm(yh ~ xh)

```

```
7 c(beta_true = 1.5, beta_hat = coef(mh)[["xh"]]) # still unbiased
```

```
## beta_true  beta_hat
##      1.500      1.604
```

Theorem 4.19 is an identity between three expressions. All three are computed below, together with the two facts the proof turns on: $\mathbf{1}'\mathbf{g} = 0$ and $\mathbf{Z}'\mathbf{g} = \mathbf{Z}'\mathbf{M}_0\mathbf{g}$.

```
1 suppressMessages({library(lmtest); library(sandwich)})
2 eh  <- residuals(mh); SSEh <- sum(eh^2)
3 s2t  <- SSEh / nh # restricted MLE, Lemma 4.14
4 g    <- eh^2 / s2t - 1
5 Z    <- cbind(xh); M0 <- diag(nh) - matrix(1/nh, nh, nh)
6 c(one_prime_g = sum(g), # = 0
7   Ztg_eq_ZtM0g = max(abs(t(Z) %*% g - t(Z) %*% M0 %*% g)))
```

```
## one_prime_g Ztg_eq_ZtM0g
##      3.475e-14      1.137e-13
```

```
1 LM_quad <- 0.5 * as.numeric(t(t(Z) %*% g) %*% solve(t(Z) %*% M0 %*% Z) %*% (t(Z) %*% g))
2 aux     <- lm(g ~ Z)
3 LM_SSR  <- sum((fitted(aux) - mean(g))^2) / 2 # SSR_z / 2
```

```

4 c(quadratic_form = LM_quad, SSR_z_over_2 = LM_SSR,
5   bptest_orig    = bptest(mh, studentize = FALSE)$statistic[["BP"]],
6   p_value        = pchisq(LM_quad, df = ncol(Z), lower.tail = FALSE))

```

```

## quadratic_form    SSR_z_over_2    bptest_orig    p_value
##      8.826e+01      8.826e+01      8.826e+01      5.741e-21

```

The statistic does not depend on h : the same LM arises whether the variance model is $\sigma_i^2 = \alpha_0 + \mathbf{z}_i' \boldsymbol{\gamma}$, $\exp(\alpha_0 + \mathbf{z}_i' \boldsymbol{\gamma})$, or any other differentiable h , because h'_0 cancelled in Step 2 of the proof. Nothing in the computation above refers to h at all.

Koenker's studentised statistic of Theorem 4.21, and the two identities its proof establishes:

```

1 v      <- eh^2
2 auxv   <- lm(v ~ Z)
3 SSRv   <- sum((fitted(auxv) - mean(v))^2); SSTv <- sum((v - mean(v))^2)
4 tau2   <- SSTv / nh                                # estimate of var(eps^2)
5 c(n_R2_z      = nh * SSRv / SSTv,
6   LM_stud_defn = (s2t^2 / tau2) * 2 * LM_quad,
7   bptest_default = bptest(mh)$statistic[["BP"]],
8   SSR_z        = 2 * LM_SSR, SSR_v_over_s2t4 = SSRv / s2t^2)

```

```
##          n_R2_z      LM_stud_defn  bptest_default
##          48.63          48.63          48.63
##          SSR_z  SSR_v_over_s2t4
##          176.52          176.52
```

```
1  c(ratio_observed = LM_quad / (nh * SSRv / SSTv),
2    tau2_over_2s4   = tau2 / (2 * s2t^2),           # equation (14)
3    tau2_over_s4    = tau2 / s2t^2,                 # = 2 under normality
4    p_orig = bptest(mh, studentize = FALSE)$p.value,
5    p_stud = bptest(mh)$p.value)
```

```
## ratio_observed  tau2_over_2s4  tau2_over_s4  p_orig.BP
##      1.815e+00      1.815e+00      3.630e+00  5.741e-21
##      p_stud.BP
##      3.088e-12
```

Both reject overwhelmingly, but they are not the same statistic. The ratio is $\hat{\tau}^2/(2\tilde{\sigma}^4)$ exactly, as (21) requires, and $\hat{\tau}^2/\tilde{\sigma}^4$ is well above the value 2 that Lemma 4.15 gives under normality: the squared residuals are more dispersed than a normal error would produce, which is precisely the circumstance in which the studentised form is the one to report.

Weighted least squares with the true weights, three ways: the closed form (11), `lm(weights=)`, and ordinary least squares on the $\sqrt{w}$ -rescaled data of Theorem 4.12:

```

1 Xh <- cbind(1, xh)
2 w  <- 1 / sig^2
3 b_wls <- drop(solve(t(Xh) %*% diag(w) %*% Xh, t(Xh) %*% diag(w) %*% yh))
4 m_wls <- lm(yh ~ xh, weights = w)
5 Xs <- sqrt(w) * Xh; ys <- sqrt(w) * yh
6 b_scaled <- drop(solve(t(Xs) %*% Xs, t(Xs) %*% ys))
7 rbind(closed_form = b_wls, lm_weights = coef(m_wls), rescaled_OLS = b_scaled)

```

```

##                xh
## closed_form  1.851 1.588
## lm_weights   1.851 1.588
## rescaled_OLS 1.851 1.588

```

```

1 V_hc1 <- vcovHC(mh, type = "HC1")
2 rbind(OLS = coef(summary(mh))[, "Std. Error"],
3       HC1 = sqrt(diag(V_hc1)),
4       WLS = coef(summary(m_wls))[, "Std. Error"])

```

```

##      (Intercept)      xh

```

```
## OLS      0.3748 0.06157
## HC1      0.2817 0.06797
## WLS      0.1558 0.04980
```

The WLS standard errors are the smallest, as Theorem 4.12 requires; OLS with HC1 is honest about its uncertainty but does not recover the lost efficiency.

7 Pitfalls

7.1 Deleting an observation because it is influential

Influence is a reason to investigate, not to delete: an unusual observation may be the most informative one.

```
1 i_max <- which.max(D_man)
2 m_drop <- lm(medv ~ lstat + age + rm, data = Boston[-i_max, ])
3 rbind(with = coef(fit), without = coef(m_drop),
4       pct_change = 100 * (coef(m_drop) - coef(fit)) / abs(coef(fit)))
```

```
##           (Intercept)    lstat          age      rm
## with           -1.175 -0.6685   0.0090913 5.019
## without         -3.997 -0.6132   0.0008842 5.437
```

```
## pct_change    -240.069   8.2683 -90.2739596  8.331
```

Diagnostic. Report the fit with and without the point and let the size of the change speak. **Fix.** Delete only for a documented reason, such as a recording error or a unit that does not belong to the population of interest.

7.2 Treating a high-leverage point as an error

Leverage depends on $\mathbf{X}$ alone, so a high-leverage point may be perfectly well fitted and carry no influence at all.

```
1 lev_top <- order(h, decreasing = TRUE)[1:5]
2 round(cbind(leverage = h[lev_top], studentised = t_man[lev_top],
3             cooks_D = D_man[lev_top]), 4)
```

```
##      leverage studentised cooks_D
## 366    0.0771         2.791  0.1605
## 215    0.0543         3.255  0.1491
## 368    0.0486         2.395  0.0725
## 369    0.0479         5.216  0.3253
## 365    0.0320        -3.374  0.0923
```

Diagnostic. Large h_{ii} with small $|t_i|$ and small D_i . **Fix.** Judge influence by D_i or DFFITS, which

combine both ingredients (Theorem 4.5).

7.3 Reading collinearity off pairwise correlations

A variable can be nearly a linear combination of several others while correlating weakly with each.

```
1 set.seed(3)
2 z1 <- rnorm(200); z2 <- rnorm(200); z3 <- z1 + z2 + rnorm(200, sd = 0.05)
3 yz <- z1 + z2 + rnorm(200)
4 mz <- lm(yz ~ z1 + z2 + z3)
5 c(cor_z1_z3 = cor(z1, z3), cor_z2_z3 = cor(z2, z3),
6   max_pairwise = max(abs(cor(cbind(z1, z2, z3))[upper.tri(diag(3))]))))
```

```
##      cor_z1_z3      cor_z2_z3 max_pairwise
##      0.6654      0.7035      0.7035
```

```
1 vif(mz)
```

```
##      z1      z2      z3
## 377.4 416.5 744.5
```

Diagnostic. Pairwise correlations near 0.7 but VIFs in the hundreds. **Fix.** VIF, which regresses each predictor on **all** the others.

7.4 Ignoring heteroscedasticity because the coefficients look fine

The point estimates are unbiased (Theorem 4.10); it is the standard errors, and therefore every t , p and interval, that are wrong.

```
1 ci_ols <- confint(mh)["xh", ]
2 ci_hc1 <- coefci(mh, vcov. = vcovHC(mh, type = "HC1"))["xh", ]
3 rbind(OLS = ci_ols, HC1 = ci_hc1,
4       width = c(diff(ci_ols), diff(ci_hc1)))
```

```
##           2.5 % 97.5 %
## OLS      1.4824 1.7247
## HC1      1.4698 1.7373
## width    0.2423 0.2675
```

Diagnostic. Compare the two sets of standard errors; a material gap means the usual formula is not to be trusted. **Fix.** Report HC1, or model the variance and use weighted least squares.

8 Exercises

Exercise 4.1. Explain why raw residuals are not comparable across observations, and why the studentised residual, not the standardised one, is used to test whether an observation is an outlier.

Solution. By the residual-variance theorem of Unit 2, $\text{var}(e_i) = \sigma^2(1 - h_{ii})$, which varies with i : a high-leverage point has a residual that is small by construction, so comparing raw residuals confuses a well-fitted point with a high-leverage one.

Standardising by $s\sqrt{1 - h_{ii}}$ equalises the variances but leaves e_i in both numerator and denominator, since observation i contributes to s . A genuine outlier inflates s and so shrinks its own r_i , masking itself. The studentised residual uses $s_{(i)}$, computed without observation i , so the observation affects only the numerator; it has an exact t_{n-k-2} distribution and is the correct test statistic.



Exercise 4.2. A regression has $n = 100$, $k = 4$, $s = 3.2$. An observation has $e_i = 9.1$ and $h_{ii} = 0.22$. Compute its standardised residual and Cook's distance, and say whether each exceeds its conventional flag.

Solution.

$$r_i = \frac{9.1}{3.2\sqrt{1-0.22}} = \frac{9.1}{3.2(0.8832)} = 3.22,$$

which exceeds 2: an outlier by the usual convention. By (5),

$$D_i = \frac{r_i^2}{k+1} \cdot \frac{h_{ii}}{1-h_{ii}} = \frac{10.37}{5} \cdot \frac{0.22}{0.78} = 2.074(0.2821) = 0.585.$$

Against the flag $D_i > 1$ this is not influential; against the more sensitive $4/n = 0.04$ it is. Its leverage 0.22 also exceeds $2(k+1)/n = 0.10$, so the observation is unusual in both respects and worth examining. ■

Exercise 4.3. Derive $y_i - \mathbf{x}_i' \hat{\boldsymbol{\beta}}^{(-i)} = e_i / (1 - h_{ii})$ from the deletion identity, and use it to explain why $\text{SSE}_{(i)} = \text{SSE} - e_i^2 / (1 - h_{ii})$.

Solution. From (3), $\mathbf{x}'_i(\hat{\boldsymbol{\beta}} - \hat{\boldsymbol{\beta}}^{(-i)}) = h_{ii}e_i/(1 - h_{ii})$, so

$$y_i - \mathbf{x}'_i\hat{\boldsymbol{\beta}}^{(-i)} = e_i + \frac{h_{ii}e_i}{1 - h_{ii}} = e_i\left(1 + \frac{h_{ii}}{1 - h_{ii}}\right) = \frac{e_i}{1 - h_{ii}}.$$

For the second claim, the deleted fit minimises the sum of squares over the remaining $n - 1$ observations. Its residual sum of squares is the full-sample SSE less the contribution of observation i , adjusted for the fact that the remaining fit no longer accommodates it; the algebra gives exactly $\text{SSE} - e_i^2/(1 - h_{ii})$, which is what makes Theorem 4.2 computable without refitting. ■

Exercise 4.4. In a model with four predictors, regressing x_2 on the other three gives $R_2^2 = 0.94$. Compute VIF_2 , and state the factor by which collinearity has widened the confidence interval for β_2 relative to an orthogonal design.

Solution. $\text{VIF}_2 = 1/(1 - 0.94) = 16.7$, above the conventional flag of 10.

By (8) the standard error carries $\sqrt{\text{VIF}_2}$, so the interval is $\sqrt{16.7} = 4.08$ times wider than it would be were x_2 orthogonal to the others. The point estimate is unaffected: collinearity costs precision,

not unbiasedness. ■

Exercise 4.5. Two predictors have $VIF = 30$ each; both t statistics are below 1, yet the overall F is 85 on 2 and 200 degrees of freedom. Is this a contradiction? What would you do?

Solution. Not a contradiction. The individual t tests ask whether each variable adds anything **given the other**, and with $VIF = 30$ the two carry nearly the same information, so neither adds much once the other is present. The F asks whether the pair together adds anything, and it plainly does. The estimates are individually imprecise and strongly correlated with each other. Options: keep both, since prediction is unharmed and the pair is jointly significant; drop one, if they measure the same construct; combine them into a single index; or shrink with ridge regression (Unit 6). What is not defensible is concluding from the t statistics that neither variable matters. ■

Exercise 4.6. Under heteroscedasticity, which of the following remain valid: unbiasedness of $\hat{\beta}$; $\text{var}(\hat{\beta}) = \sigma^2(\mathbf{X}'\mathbf{X})^{-1}$; the Gauss–Markov theorem; the t distribution of the usual statistic?

Solution. Only unbiasedness. Its proof used A1–A2 alone (Theorem 4.10).

The variance formula fails, being replaced by the sandwich (9). Gauss–Markov fails, since its proof used $\text{var}(\boldsymbol{\varepsilon}) = \sigma^2 \mathbf{I}$; weighted least squares is the efficient estimator instead (Theorem 4.12). The usual t statistic fails because its denominator estimates the wrong variance, so its p -values are wrong in an unknown direction. HC1 standard errors repair this without touching the estimates.



Exercise 4.7. Show that weighted least squares with $w_i = 1/\sigma_i^2$ is ordinary least squares on the data $(y_i/\sigma_i, \mathbf{x}_i/\sigma_i)$, and explain the weighting intuitively.

Solution. The transformation is Theorem 4.12: with $\mathbf{W}^{1/2} = \text{diag}(1/\sigma_i)$, the starred model has $\text{var}(\boldsymbol{\varepsilon}^*) = \mathbf{I}$ and OLS on it gives $(\mathbf{X}'\mathbf{W}\mathbf{X})^{-1}\mathbf{X}'\mathbf{W}\mathbf{Y}$, which is $\hat{\boldsymbol{\beta}}_W$.

Intuitively, dividing observation i by σ_i puts every observation on a common noise scale. An observation with large σ_i says little about the regression line and receives a small weight $1/\sigma_i^2$; a precise observation receives a large one. Least squares weights all observations equally, which is right only when they are equally informative.



Exercise 4.8. For each symptom, name the assumption at risk and one remedy: (a) residuals curve as a function of the fitted values; (b) residual spread widens with the fitted value; (c) one observation has $D_i = 4$; (d) two predictors have $VIF = 45$.

Solution. (a) [A1](#), linearity. Add a quadratic term or transform a variable (Unit 3 §3.4).
(b) [A3](#), constant variance. Report HC1 standard errors, use weighted least squares, or transform the response (§5).
(c) No assumption. Influence is not a violation; investigate the observation, report the fit with and without it, and delete only for a documented reason.
(d) [A4](#) is not violated, $\mathbf{X}'\mathbf{X}$ being still invertible, but it is nearly so. Drop or combine one of the pair, or shrink (Unit 6). ■

Exercise 4.9. In R: (a) verify Cook's distance from ([5](#)) against `cooks.distance`; (b) confirm the deletion identity against explicit refits; (c) compare OLS, HC1 and WLS standard errors on heteroscedastic data.

Solution. (a) is §6.2, (b) the first chunk there, (c) §6.4. In (c), weighted least squares is smallest on both coefficients, as Theorem 4.12 requires, because it is the only one of the three that uses the true σ_i^2 . HC1 and OLS do not order each other: HC1 is larger than OLS on the slope and smaller on the intercept, which is the point of it. HC1 is not an inflated OLS standard error but an estimate of a different quantity, the true sampling variability, and that can lie on either side of the one the constant-variance formula reports. ■